

E-Commerce Sales & Customer Behaviour Analysis

SQL Server + Power BI

1. Project Overview

This project analyzes sales performance and customer purchasing behaviour for an online retail business using SQL Server and Power BI. The analysis uses the UCI Online Retail dataset, containing transactional data from a UK-based online retailer between December 2010 and December 2011. The project focuses on revenue performance, customer behaviour, product performance, geographic sales distribution, purchasing frequency, customer value, cancellations, and customer segmentation.

2. Business Problem

An e-commerce business needs to understand:

- How much revenue is being generated?
- How are sales changing over time?
- Which countries contribute the most revenue?
- Which products generate the highest revenue and sales volume?
- Which customers are the most valuable?
- How frequently do customers purchase?
- How many customers are repeat customers?
- Which customers belong to different value levels?
- How significant are cancellations?
- Which customer segments should receive greater attention?

The objective was to answer these questions using transactional data and convert the findings into actionable business insights.

3. Project Objectives

- Clean and validate the raw transactional dataset.
- Calculate key sales and customer KPIs.
- Analyze monthly sales performance.
- Identify top-performing countries and products.
- Analyze customer purchasing behaviour.
- Identify repeat and one-time customers.
- Segment customers based on their value.
- Perform RFM analysis.
- Analyze cancellations separately from completed sales.
- Build an interactive Power BI dashboard for business reporting.

4. Dataset

Dataset: UCI Online Retail Dataset. The dataset contains 541,909 raw transaction records.

Main columns:

- InvoiceNo — Invoice/order number
- StockCode — Product code
- Description — Product description
- Quantity — Number of units purchased
- InvoiceDate — Transaction date and time
- UnitPrice — Price per unit
- CustomerID — Customer identifier
- Country — Customer's country

Invoice numbers beginning with C represent cancellations. The dataset covers transactions from 01 December 2010 to 09 December 2011.

5. Tools & Technologies

SQL Server / SSMS

- Data cleaning and validation
- Duplicate removal
- KPI calculations
- Sales, customer, product and country analysis
- Monthly trend analysis
- RFM analysis
- Cancellation analysis
- Customer segmentation

Power BI

- Data visualization
- DAX measures
- KPI cards
- Interactive charts
- Customer segmentation
- Dashboard design
- Business reporting

6. Data Cleaning & Preparation

The original dataset was preserved as `dbo.OnlineRetail_Raw`. A separate cleaned table, `dbo.OnlineRetail_Cleaned`, was created for analysis.

- Removed records with missing product descriptions.
- Removed transactions where Quantity ≤ 0 .
- Removed transactions where UnitPrice ≤ 0 .
- Removed records with missing CustomerID.
- Removed exact duplicate transactions.

- Created a calculated SalesAmount field.

$\text{SalesAmount} = \text{Quantity} \times \text{UnitPrice}$

Raw dataset: 541,909 rows. Final analytical dataset after cleaning and duplicate removal: 392,692 rows.

Duplicate validation confirmed no remaining exact duplicate groups. Cancellation transactions were analyzed separately and excluded from the main sales analysis.

7. Overall Sales Performance

KPI	Result
Total Revenue	8,887,208.89
Total Orders	18,532
Total Customers	4,338
Total Units Sold	5,152,002
Average Order Value	479.56

These KPIs were used as the main performance indicators in the Power BI dashboard.

8. Monthly Sales Analysis

- November 2011 generated the highest full-month revenue.
- November revenue reached approximately 1.16 million.
- September 2011 recorded the strongest month-over-month revenue growth at approximately 47.61%.
- December 2011 was incomplete, with data available only through 09 December, so it was not interpreted as a complete-month decline.

9. Country Analysis

The United Kingdom is the dominant revenue market.

Country	Revenue
United Kingdom	7,285,024.64
Netherlands	285,446.34
EIRE	265,262.46
Germany	228,678.40
France	208,934.31

The UK represents the dominant revenue market, while the Netherlands, EIRE, Germany, and France form important secondary markets.

10. Product Analysis

Product performance was analyzed using total revenue, quantity sold, and average unit price.

Highest-revenue product: PAPER CRAFT, LITTLE BIRDIE

- Revenue: 168,469.60

- Units sold: 80,995

Analyzing both revenue and quantity is important because high sales volume does not necessarily mean the highest revenue contribution.

11. Customer Behaviour Analysis

- Total customers: 4,338
- One-time customers: 1,493
- Repeat customers: 2,845
- Repeat customer rate: approximately 66%
- Average orders per customer: 4.27
- Highest number of orders by one customer: 209
- Average revenue per customer: 2,048.69
- Highest customer revenue: 280,206.02

The majority of customers were repeat purchasers, indicating relatively strong repeat-purchase behaviour. Customer value was highly concentrated among a smaller group.

12. Customer Value Segmentation

Segment	Customers	Revenue
Low Value	1,764	456,767.47
Medium Value	2,300	3,639,787.09
High Value	274	4,790,654.33

High Value customers represented only around 6.3% of customers but contributed approximately 54% of total revenue. This demonstrates significant revenue concentration among a relatively small group of customers.

13. RFM Analysis

RFM analysis was performed to better understand customer behaviour.

- Recency — How recently a customer purchased
- Frequency — How often a customer purchased
- Monetary — How much revenue the customer generated

SQL techniques used included CTEs, NTILE(), window functions, aggregations, CASE expressions, and subqueries. Customers were classified into groups such as Best Customers, Loyal Customers, At Risk, and Other Customers.

14. Cancellation Analysis

Cancellations were analyzed separately from completed sales.

Metric	Result
Cancellation Orders	3,836
Cancellation Rows	9,288

Cancelled Units	277,574
Cancelled Value	896,812.49

Cancellation transactions were excluded from the cleaned sales dataset to prevent them from artificially reducing sales performance metrics.

15. Power BI Dashboard

The final Power BI dashboard consists of two pages.

Page 1 — Executive Overview

- KPI Cards: Total Revenue, Total Orders, Total Customers, Average Order Value
- Monthly Sales Trend — shows revenue performance over time.
- Top 10 Countries by Revenue — highlights the most important geographic markets.

Page 2 — Product & Customer Analysis

- Top 10 Products by Revenue
- Top 10 Products by Quantity Sold
- Top 10 Customers by Revenue
- Top 10 Customers by Orders
- Customer Value Segmentation

16. Key Business Insights

Revenue is heavily concentrated in the UK: The UK generated approximately 7.29 million in revenue, significantly exceeding other countries.

A small group of customers drives a large share of revenue: Only around 6.3% of customers fall into the High Value segment, yet they contribute approximately 54% of revenue.

Repeat customers form the majority: Approximately 66% of customers are repeat customers, indicating strong repeat-purchase behaviour.

November was the strongest full month: November 2011 generated approximately 1.16 million in revenue.

September experienced strong growth: September 2011 recorded approximately 47.61% month-over-month revenue growth.

Product volume and revenue tell different stories: Products with the highest unit volume are not necessarily the products generating the highest revenue.

Cancellations represent significant business activity: Cancelled transactions represented approximately 896.8K in value.

17. Business Recommendations

Focus on high-value customers

Develop loyalty programs, personalized offers, and targeted communication for high-value customers.

Improve customer retention

Encourage repeat purchases through personalized recommendations, promotions, and customer engagement strategies.

Investigate cancellation behaviour

Analyze cancelled products, customers, and time periods to identify recurring causes of cancellations and reduce potential revenue leakage.

Strengthen international markets

The UK dominates revenue, but the Netherlands, EIRE, Germany, and France show meaningful revenue potential.

Monitor product performance

Track both product revenue and quantity sold to identify high-volume products, high-revenue products, and pricing or promotional opportunities.

18. Conclusion

This project demonstrates how transactional e-commerce data can be transformed into meaningful business insights using SQL Server and Power BI. SQL Server was used to clean, validate, transform, and analyze the data, while Power BI was used to communicate the results through an interactive two-page dashboard.

The analysis revealed strong repeat-customer behaviour, significant revenue concentration among high-value customers, strong UK market dominance, important product-level differences, and substantial cancellation activity.

Overall, the project demonstrates practical skills in data cleaning, SQL analysis, customer analytics, RFM segmentation, business intelligence, data visualization, and insight generation.

19. Skills Demonstrated

SQL Server / SSMS

- Data cleaning
- Data validation
- Duplicate handling
- Aggregations
- CTEs
- Subqueries
- CASE
- LAG()
- NTILE()

- Window functions
- Customer segmentation
- RFM analysis

Power BI

- Data modelling
- DAX measures
- Calculated columns
- KPI cards
- Bar charts
- Line charts
- Donut charts
- Top N filtering
- Dashboard design
- Data visualization
- Business reporting

Business Analytics

- Sales performance analysis
- Customer behaviour analysis
- Product analysis
- Geographic analysis
- Customer segmentation
- Retention analysis
- Cancellation analysis
- Business recommendations

20. Project Result

SQL Server → Clean & Analyze → Power BI → Visualize → Business Insights

This project provides an end-to-end example of using transactional data to support sales performance monitoring, customer understanding, and business decision-making.